

SQA / Software Testing

Login Module — 25 Test Cases

Project: E-Commerce Web Application

Purpose: Manual/functional testing practice for the Login Module. The test cases are written in professional English for SQA portfolio and interview preparation.

Standard Test Case Format: Test Case ID • Test Scenario • Preconditions • Test Steps • Test Data • Expected Result • Actual Result • Status

TC_LOGIN_001 — Login with valid email and valid password

Test Case ID: TC_LOGIN_001

Test Scenario: Login with valid email and valid password

Preconditions: User is on the login page.

Test Steps:

1. Enter a valid email.
2. Enter a valid password.
3. Click the Login button.

Test Data: Email: testuser@gmail.com

Password: Valid@123

Expected Result: User should be logged in successfully and the dashboard/home page should open.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_002 — Login with invalid email and invalid password

Test Case ID: TC_LOGIN_002

Test Scenario: Login with invalid email and invalid password

Preconditions: User is on the login page.

Test Steps:

1. Enter an invalid email.
2. Enter an invalid password.
3. Click the Login button.

Test Data: Email: wrong@gmail.com

Password: Wrong@123

Expected Result: Login should fail and an appropriate error message should be displayed, e.g., "Invalid email or password."

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_003 — Login with valid email and invalid password

Test Case ID: TC_LOGIN_003

Test Scenario: Login with valid email and invalid password

Preconditions: User is on the login page.

Test Steps:

1. Enter a valid email.
2. Enter an incorrect password.
3. Click the Login button.

Test Data: Email: testuser@gmail.com

Password: Wrong@123

Expected Result: Login should fail and an appropriate error message should be displayed.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_004 — Login with valid email and empty password

Test Case ID: TC_LOGIN_004

Test Scenario: Login with valid email and empty password

Preconditions: User is on the login page.

Test Steps:

1. Enter a valid email.
2. Leave the password field empty.
3. Click the Login button.

Test Data: Email: testuser@gmail.com

Password: Empty

Expected Result: Login should fail and an appropriate validation message should be displayed below the password field, e.g., "Password is required."

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_005 — Login with empty email and valid password

Test Case ID: TC_LOGIN_005

Test Scenario: Login with empty email and valid password

Preconditions: User is on the login page.

Test Steps:

1. Leave the email field empty.
2. Enter a valid password.
3. Click the Login button.

Test Data: Email: Empty

Password: Valid@123

Expected Result: Login should fail and an appropriate validation message should be displayed below the email field, e.g., "Email is required."

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_006 — Login with both email and password empty

Test Case ID: TC_LOGIN_006

Test Scenario: Login with both email and password empty

Preconditions: User is on the login page.

Test Steps:

1. Leave the email field empty.
2. Leave the password field empty.
3. Click the Login button.

Test Data: Email: Empty

Password: Empty

Expected Result: Login should fail and appropriate validation messages should be displayed for both email and password fields.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_007 — Login with invalid email format

Test Case ID: TC_LOGIN_007

Test Scenario: Login with invalid email format

Preconditions: User is on the login page.

Test Steps:

1. Enter an email with an invalid format, e.g., testuser@.
2. Enter a valid password.
3. Click the Login button.

Test Data: Email: testuser@

Password: Valid@123

Expected Result: Login should fail and an appropriate email validation message should be displayed, e.g., "Please enter a valid email address."

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_008 — Login with leading/trailing spaces in email

Test Case ID: TC_LOGIN_008

Test Scenario: Login with leading/trailing spaces in email

Preconditions: User is on the login page.

Test Steps:

1. Enter a valid email with leading/trailing spaces.
2. Enter a valid password.
3. Click the Login button.

Test Data: Email: testuser@gmail.com

Password: Valid@123

Expected Result: The application should properly handle unnecessary spaces in the email. If the email is valid, the expected login behavior should occur.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_009 — Password visibility / eye icon

Test Case ID: TC_LOGIN_009

Test Scenario: Password visibility / eye icon

Preconditions: User is on the login page.

Test Steps:

1. Enter a password.
2. Click the eye icon.
3. Click the eye icon again.

Test Data: Password: Valid@123

Expected Result: On the first click, the password should become visible; on the second click, it should become masked/hidden again.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_010 — Login button with valid credentials

Test Case ID: TC_LOGIN_010

Test Scenario: Login button with valid credentials

Preconditions: A valid registered account is available.

Test Steps:

1. Enter a valid email.
2. Enter a valid password.
3. Click the Login button.

Test Data: Valid email + Valid password

Expected Result: The Login button should submit the request and successful login should occur.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_011 — Login button with required fields empty

Test Case ID: TC_LOGIN_011

Test Scenario: Login button with required fields empty

Preconditions: User is on the login page.

Test Steps:

1. Keep the email and password fields empty.
2. Click the Login button.

Test Data: Email: Empty

Password: Empty

Expected Result: Invalid submission should be prevented and required-field validation messages should be displayed.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_012 — Email field accepts valid email length

Test Case ID: TC_LOGIN_012

Test Scenario: Email field accepts valid email length

Preconditions: User is on the login page.

Test Steps:

1. Enter a valid email within the supported length.
2. Enter a valid password.
3. Click the Login button.

Test Data: A valid email within supported length

Password: Valid@123

Expected Result: The valid email should be accepted and normal authentication should proceed.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_013 — Password field masks typed characters

Test Case ID: TC_LOGIN_013

Test Scenario: Password field masks typed characters

Preconditions: User is on the login page.

Test Steps:

1. Select the password field.
2. Type a password.

Test Data: Password: Valid@123

Expected Result: Password characters should be masked/hidden by default while typing.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_014 — Login page validation / UI coverage

Test Case ID: TC_LOGIN_014

Test Scenario: Login page validation / UI coverage

Preconditions: User is on the login page.

Test Steps:

1. Inspect the login page fields and buttons.
2. Check required validations and UI behavior against the requirements.

Test Data: As per application requirements

Expected Result: The login page UI and validation should work properly according to the requirements/documentation.

Actual Result: Pass / Fail

Status: Completed

TC_LOGIN_015 — Login with email containing internal spaces

Test Case ID: TC_LOGIN_015

Test Scenario: Login with email containing internal spaces

Preconditions: User is on the login page.

Test Steps:

1. Enter an invalid email containing an internal space.
2. Enter a valid password.
3. Click the Login button.

Test Data: Email: test user@gmail.com

Password: Valid@123

Expected Result: The invalid email should be rejected and a clear validation message should be displayed.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_016 — Password minimum boundary length

Test Case ID: TC_LOGIN_016

Test Scenario: Password minimum boundary length

Preconditions: The password minimum length requirement is known.

Test Steps:

1. Enter a password exactly one character shorter than the minimum allowed length.
2. Enter a valid email.
3. Click the Login button.

Test Data: Password: One character less than minimum

Expected Result: If minimum length is enforced, a validation message should be displayed and the invalid authentication request should be prevented.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_017 — Password minimum valid boundary

Test Case ID: TC_LOGIN_017

Test Scenario: Password minimum valid boundary

Preconditions: The password minimum length requirement is known.

Test Steps:

1. Enter a password with exactly the minimum allowed length.
2. Enter a valid email.
3. Click the Login button.

Test Data: Password: Exactly minimum length

Expected Result: A password meeting the minimum allowed length should be accepted and authentication should behave according to the requirements.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_018 — Password maximum boundary

Test Case ID: TC_LOGIN_018

Test Scenario: Password maximum boundary

Preconditions: The password maximum length requirement is known.

Test Steps:

1. Enter a password with the maximum supported length.
2. Attempt to log in.

Test Data: Password: Maximum supported length

Expected Result: The application should correctly handle the maximum supported length without unexpected truncation or errors.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_019 — Very long password input

Test Case ID: TC_LOGIN_019

Test Scenario: Very long password input

Preconditions: User is on the login page.

Test Steps:

1. Type or paste a very long password.
2. Click the Login button.

Test Data: Password: Very long string

Expected Result: The application should handle the input safely; the UI should not break or crash, and an appropriate validation/authentication response should be provided.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_020 — Password special characters

Test Case ID: TC_LOGIN_020

Test Scenario: Password special characters

Preconditions: A valid account is available.

Test Steps:

1. Use supported special characters in the password.
2. Click the Login button.

Test Data: Password containing: @ # \$ % &

Expected Result: Supported special characters should be accepted correctly and, with valid credentials, expected authentication should occur.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_021 — Email case handling

Test Case ID: TC_LOGIN_021

Test Scenario: Email case handling

Preconditions: A registered email is known.

Test Steps:

1. Enter the same email using different uppercase/lowercase combinations.
2. Enter a valid password.

3. Click the Login button.

Test Data: Email: TESTUSER@GMAIL.COM

Password: Valid@123

Expected Result: The application should handle email casing consistently according to the requirements and should not unexpectedly reject a valid account.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_022 — Multiple failed login attempts

Test Case ID: TC_LOGIN_022

Test Scenario: Multiple failed login attempts

Preconditions: User is on the login page.

Test Steps:

1. Attempt to log in multiple times using an incorrect password.
2. Observe the response after each attempt.

Test Data: Same valid email + wrong password

Expected Result: The application should follow the defined failed-attempt behavior, such as an appropriate error, rate limiting, temporary lockout, or CAPTCHA if required.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_023 — Forgot Password navigation

Test Case ID: TC_LOGIN_023

Test Scenario: Forgot Password navigation

Preconditions: User is on the login page.

Test Steps:

1. Click the Forgot Password link.

Test Data: N/A

Expected Result: The user should be correctly navigated to the password recovery/reset page or process.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_024 — Password case sensitivity

Test Case ID: TC_LOGIN_024

Test Scenario: Password case sensitivity

Preconditions: A registered account is available.

Test Steps:

1. Enter the registered email.
2. Enter the password with changed uppercase/lowercase letters.
3. Click the Login button.

Test Data: Email: testuser@gmail.com
Registered Password: Valid@123
Entered Password: valid@123

Expected Result: Login should not be successful because passwords are case-sensitive.

Actual Result: Pass / Fail

Status: Planned

TC_LOGIN_025 — Login after successful authentication

Test Case ID: TC_LOGIN_025

Test Scenario: Login after successful authentication

Preconditions: A valid account is available.

Test Steps:

1. Log in using valid credentials.
2. After the dashboard opens, check browser back/refresh behavior.
3. Observe access to the protected page.

Test Data: Valid email + Valid@123

Expected Result: The authenticated user should receive expected access to the protected area, and sensitive/protected data should not be exposed in an unauthorized state.

Actual Result: Pass / Fail

Status: Planned

Interview Quick Revision

What is a test case?

A test case is a detailed set of preconditions, steps, test data, expected result, actual result, and status used to verify a specific behavior.

Scenario vs Test Case

A scenario explains WHAT to test; a test case explains HOW to test it.

Positive Testing

Checking expected successful behavior using valid input.

Negative Testing

Checking application behavior against invalid, missing, unexpected, or boundary inputs.

Why test empty fields?

To verify that required-field validation works correctly.

Why test boundary values?

To find defects around minimum and maximum limits.

What is a bug?

A mismatch or defect between expected behavior and actual behavior.

When is Actual Result filled?

After observing the actual application behavior during manual testing.

If login fails with valid credentials?

Reproduce the issue, capture evidence, compare the behavior with requirements, and create a clear bug report containing steps, expected vs. actual result, severity, and priority.

Next SQA Stage: Manually execute these 25 test cases, record Actual Result and Pass/Fail, then proceed to Bug Reporting, Smoke/Sanity/Regression Testing, Responsive/Cross-Browser Testing, Basic SQL, and Postman/REST API practice.